Weight and local gravity

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Learning target: Distinguish mass from gravitational weight.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 20.*
$a=7$ m/s² right; $v_f=28$ m/s right. Constant net force and mass throughout.

02. Worked example: explain the model
12 N and 6 N downward toward each body's center; mass unchanged at 3 kg.

03. Read and interpret the model
Mass 2 kg in both places; weights 20 N and 10 N.

04. Apply the physics
15 N; 6 kg weighs 30 N, twice as much.

05. Explain your reasoning
No. Scale measures contact/support force; shared free fall can give near-zero reading while gravity acts.

06. Review: connect an earlier idea *Review: Unit 1, Day 13.*
Net force 24 N right; acceleration 6 m/s² right.

07. Exit check
4 kg and 12 N.

08. STRETCH: test the model
No: 2 kg at 6 N/kg and 3 kg at 4 N/kg both weigh 12 N.

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01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 20.*
x interval [-55, -45] km; y interval [63, 77] km. Simplified geometry, gravity, Earth motion or missing observations limit the model. Bounds are not probabilities.

02. Worked example: explain the model
12 N and 6 N downward toward each body's center; mass unchanged at 3 kg.

03. Read and interpret the model
Mass 2 kg in both places; weights 20 N and 10 N.

04. Apply the physics
588 N; 96 N; g=240/60=4 N/kg.

05. Explain your reasoning
No. Scale measures contact/support force; shared free fall can give near-zero reading while gravity acts.

06. Review: connect an earlier idea *Review: Unit 1, Day 13.*
Net -32 N; a=-8 m/s². Need current velocity: leftward motion speeds up; rightward motion slows.

07. Exit check
Mass 5 kg; new weight 10 N.

08. STRETCH: test the model
Mass 6 kg; first g=12 N/kg. Weight changes with field, not necessarily mass.